31-07-23

RECURSION

1. Base Case → [iteration == len(array):]
2. Recursive statement

Iterate an array through recursion:

```
a = [15, 14, 33, 41, 25]

def iter_array(arr, i=0):
    if i == len(arr):
        return
    print(arr[i])
    iter_array(arr, i+1)
```

i → index i = 0	Print
0	15
1	14

Backward iteration

```
a = [15, 14, 33, 41, -25]
iter_array (a, len(a) - 1)

def iter_array (arr, i):
    if i == ~~len(arr) - 1~~ len(arr):
        return
    print (arr[i])
    iter_array ( arr, i-1)
```

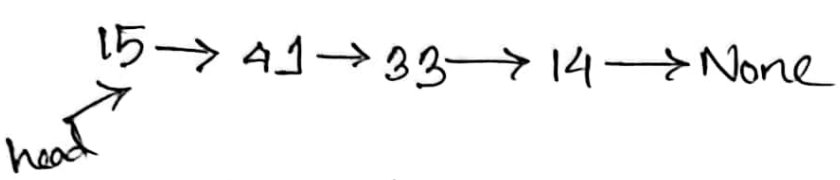
OR

```
print (arr [len(arr) - 1 - i])
iter_array (arr, i+1).
```

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Linked list iteration



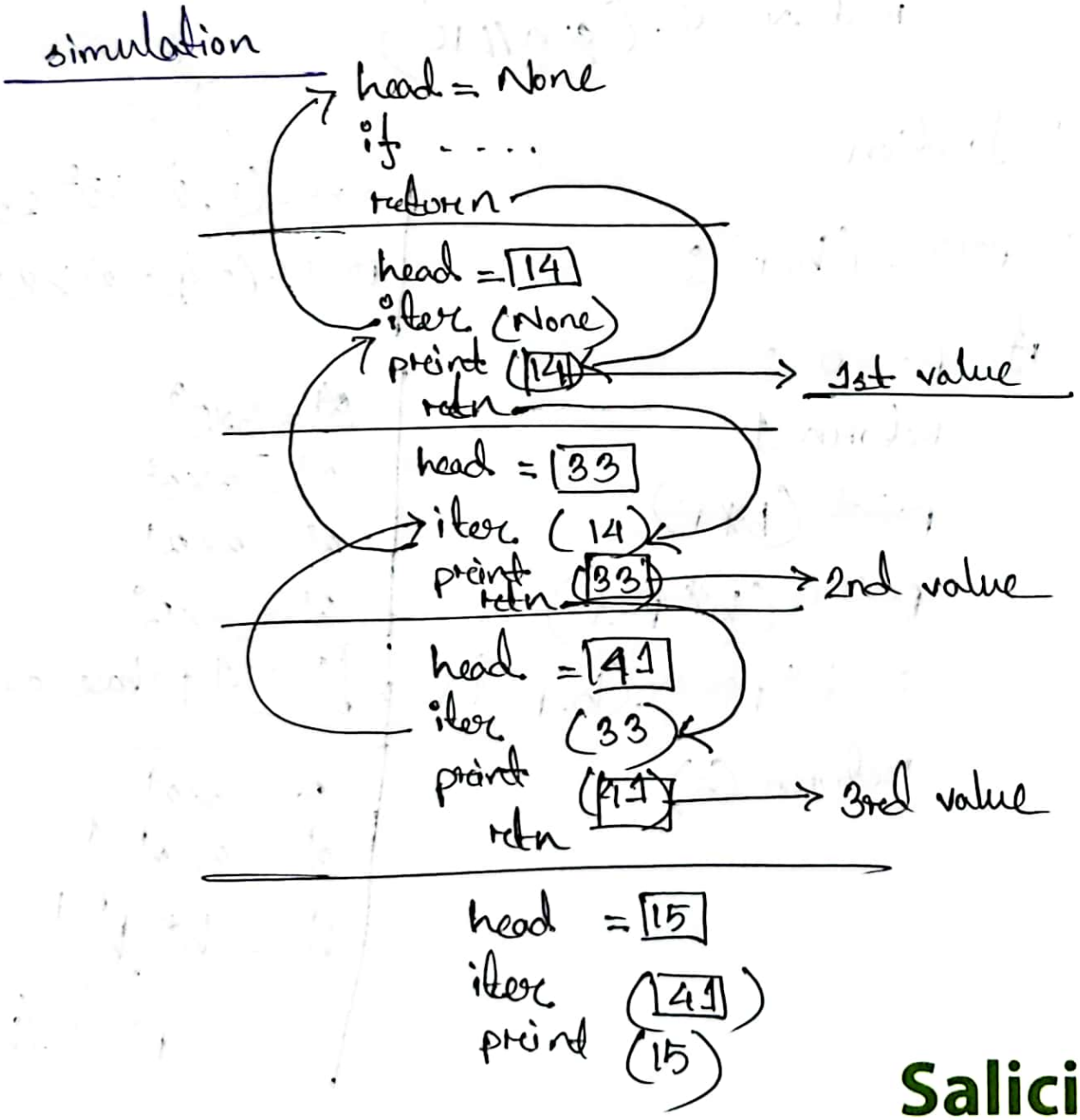

```
while head != None:
    print ...
    head = head.next
```

```
def iter_linkedlist (head):
    if head == None:
        return
    print (head.elem)
    iter_linkedlist (head.next)
```

Backward iteration:

```
def iter_linkedlist (head):
    if head == None:
        return
    ⇒ iter_linkedlist (head.next)
    ⇒ [print (head.elem)]
```

simulation



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integer
n = 9 4 3 7 | output : 7 3 4 9

```
def mod_value (n):
    if n == 0:
        return
    else sum = n%10
         print (sum)
         mod_value (n//10)
```

power function

```
def power (b, p):
    if p==0 :
        return 1
        print (b*p)
        power (b* p-1)
        a = b* power (b, p-1)
        return (a)
```

base, power

power (5, 2) => $5^2 = 25$

power (3, 4) = $3^4 \Rightarrow 81$

$3^4 = 3 \times 3^3$

$3^3 = 3 \times 3^2$

$3^2 = 3 \times 3^1$

$3^1 = 3 \times 3^0$

$\boxed{3^0 = 1}$ → base case.

$3^5 = 3 \times 3^4$

$3^p = 3 \times 3^{p-1}$

$b^p = b * b^{p-1}$

b^{p-1} → power (b, p-1)

Binary Search:

Condition: Array must be sorted!

search : 33

arr = 2 4 7 15 19 33 45

0 1 2 3 4 5 6

search space → array র কতটুকু space এ খুঁজবা হবে।

Search space

var : low (left most idx) = 0

high (right most idx) = 6

code

```
low = 0
high = len(arr) - 1
mid = (low+high) // 2
```

- if mid element > search element
  high = mid - 1
- if mid element < search element
  low = mid + 1
- mid element == search element
  return True / False.

- if low > high:
  return False

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```
def binary_search(arr, search, low, high):
    mid = low+high//2
    if arr[mid] == search
        return True
    elif low > high:
        return False
    elif arr[mid] < search:
        low = mid+1
    elif arr[mid] > search:
        high = mid-1

result=binary_search(arr, search, low, high)
return (result)
```